

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2155

Elmo 138 kV
October 19, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2155. The Interconnection Customer proposes a 1226.3 MW natural gas combined cycle facility interconnecting at the Elmo 138 kV substation in the Virginia area. The request was received on August 12, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 412 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$57,780,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Elmo-Hickory Substation 138 KV	Robinson Avenue-Elmo 138 KV line	286.5	402	88.4
Robinson Avenue-Elmo 138 KV line	Elmo-Hickory 138 KV	244.0	442	84.2
Summit Farms Solar-Robinson Avenue 138 KV line	Elmo-Hickory Substation 138 KV line	200.3	513	84.2
Hartselle-Summit Farms Solar 138 KV	Hickory-Elmo 138 KV	190.9	525	84.1
Hickory-Harbor Beach Substation 138 KV	Robinson Avenue-Elmo 138 KV L	184.5	590	88.4
No Mans Land-Ne-Hobbs 138 KV	Elmo-Robinson Avenue 138 KV	152.9	648	88.4
Ne-Hobbs-Harbor Beach Substation 138 KV	Robinson Avenue-Elmo 138 KV line	149.4	688	88.4
Harvell-Oyster River 138 KV	Haggin-North Tower Tap 345 KV	128.7	191	5.0
Oyster Riv-Cedar Falls Wi 138 KV	N Tower Tap-Haggin 345 KV	128.5	212	5.0
Wheelabrator Portsmouth-Victory Way 138 KV	Wheelabrator Portsmouth-Gosport 138 KV line	114.6	329	9.4
Thole St-Diamond Springs 138 KV	Wheelabrator Portsmouth-Victoryway 138 KV	112.3	211	5.7
Pine Bluff West-Trans Bay Cable Facility 138 KV	Avra Vly-Chuckatuck 138 KV line	109.3	237	7.4
Hartselle 345-138 kV TX	Elmo Substation-Hickory 138 KV	108.0	688	75.4
Harvell-Harrowgate 138 KV	N Tower Tap-Haggin 345 KV L	105.8	192	5.0
N Tower Tap-Mccartney 345 KV	Bear-Mecklenburg Power Sta 345 KV line	105.0	1,698	28.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Jim Hill-Haggin 138 KV	Skaggs Island Jct 2-Mitchell Dam 345 KV	103.9	413	5.6
Jim Hill-Cedar Bayou Plant 138 KV	Mitchell Dam-Skaggs Island Junction 2 345 KV line	101.9	352	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hickory Substation-Elmo 138 KV	Rebuild line with higher-capacity conductor	\$2,472,000	79% (\$1,946,000)
Robinson Avenue Substation-Elmo 138 KV L	Rebuild line with higher-capacity conductor	\$5,872,000	78% (\$4,591,000)
Robinson Avenue-Summit Farms Solar 138 KV	Rebuild line with higher-capacity conductor	\$7,959,000	78% (\$6,223,000)
Hartselle-Summit Farms Solar 138 KV	Rebuild line with higher-capacity conductor	\$5,305,000	78% (\$4,148,000)
Hickory-Harbor Beach Substation 138 KV	Rebuild line with higher-capacity conductor	\$6,174,000	79% (\$4,861,000)
No Mans Land-Ne-Hobbs 138 KV	Rebuild line with higher-capacity conductor	\$5,035,000	79% (\$3,964,000)
Ne-Hobbs-Harbor Beach 138 KV	Rebuild line with higher-capacity conductor	\$5,857,000	79% (\$4,612,000)
Harvell-Oyster River 138 KV Line	Rebuild line with higher-capacity conductor	\$2,176,000	19% (\$409,000)
Oyster River-Cedar Falls Wi 138 KV Line	Rebuild line with higher-capacity conductor	\$2,245,000	19% (\$422,000)
Wheelabrator Portsmouth-Victory Way 138 KV	Rebuild line with higher-capacity conductor	\$9,748,000	34% (\$3,356,000)
Thole St-Diamond Springs 138 KV	Rebuild line with higher-capacity conductor	\$4,125,000	38% (\$1,547,000)
Pine Bluff West-Trans Bay Cable Facility 138 KV	Reconductor line	\$1,905,000	26% (\$490,000)
Hartselle 345-138 kv TX	Replace transformer with higher-rated unit	\$17,963,000	61% (\$10,923,000)
Harvell-Harrowgate 138 KV	Reconductor line	\$1,521,000	19% (\$286,000)
N Tower Tap-Mccartney 345 KV	Reconductor line	\$52,991,000	18% (\$9,650,000)
Jim Hill-Haggin 138 KV	Reconductor line	\$468,000	16% (\$73,000)
Jim Hill-Cedar Bayou Plant 138 KV	Reconductor line	\$1,777,000	16% (\$279,000)
	Total allocated to project		\$57,780,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.